

WL stat_obs p_value IMF1 IMF2 IMF3 IMF4 0.030000 48.969698 0.000100 -0.050913
 -0.068552 -0.048556 0.061210 0.035000 48.107214 0.000100 -0.044921 -0.071515
 -0.066169 0.076311 0.040000 50.622517 0.000100 -0.039372 -0.083078 -0.090185
 -0.072064 0.045000 50.787299 0.000100 -0.045422 -0.105136 -0.110862 -0.065604
 0.050000 50.438822 0.000100 -0.053482 -0.114618 -0.129916 -0.025598 Best WL by mean |
 IMF2 & IMF3|: 0.05

ChatGPT said:

Thought for 48s

Nice sweep — this is really informative. Here’s a tight read of what your WL scan shows and what it means for the SN-IA ↔ BAO “locking” story.

Executive summary

- The locking signal for the science-relevant bands (IMF2–IMF3) **strengthens monotonically** as the $\log(1+z)$ window widens from **WL = 0.03 → 0.05**, with the **largest mean |lock| at WL = 0.05** (your “Best WL”).
- The **sign is negative** for IMF2–IMF3 across the scan, i.e. higher BAO proxy values associate with **lower instantaneous frequency** in those IMFs (anti-correlated ridge trend).
- IMF4 **changes sign** near **WL $\approx$ 0.04**, consistent with a different phase/scale relationship (or a separate component) from the IMF2–3 band.
- The IMF-energy surrogate test remains **highly significant** ($p \approx 1 \times 10^{-4}$) for all WL — the signal power in the focus IMFs is not a fluke of the sampling.
- Practical takeaway: for locking diagnostics and de-chirp work, **WL $\approx$ 0.045–0.050** is a good default; you haven’t obviously over-smoothed yet.

What the two plots + table are telling us

1) Lock *strength* vs WL (|score|)

- IMF2: 0.069 → 0.115
- IMF3: 0.049 → 0.130

Both rise smoothly with WL. In this metric (local, z-scored correlation inside a sliding window in L), a wider window reduces variance and reveals the underlying slow co-variation between the IMF ridge frequency $f(L)$ and the BAO proxy $b(L)$.

2) Lock *sign* vs WL

- IMF2, IMF3 stay **negative** and become more negative as WL grows.
- IMF4 flips (+ → −) at **WL $\approx$ 0.04** and weakens by **WL = 0.05**. That phase flip suggests IMF4 is not the same physical driver as IMF2–3 (or it mixes with one at different L).

3) Surrogate energy test

Your $\text{stat_obs} \sim 49\text{--}51$ with $p \approx 1\text{e-}4$ for every WL: there is robust, non-random power in the focus IMFs. The locking trend is therefore acting on a real mode, not noise.

Why WL matters (and why 0.05 looks right)

The lock score you're using is, in essence, an **average of local, standardized correlations**:

$$\text{lock} = \frac{1}{N_{\text{win}}} \sum_j \text{corr} \left(\frac{f(L) - \mu_{f,j}}{\sigma_{f,j}}, \frac{b(L) - \mu_{b,j}}{\sigma_{b,j}} \right),$$

where windows of half-width $\frac{1}{2} \text{WL}$ slide along $L = \log(1+z)$.

- **Too narrow WL** → noisy correlation estimates.
- **Too wide WL** → you might smear genuine L-dependent sign changes.

Your curves keep **gaining** strength through 0.05 with **consistent sign**, which means you're not yet smearing away structure. It's a good operating point. (I'd still do a quick 0.055–0.07 check to confirm you hit a plateau rather than a continued rise.)

Relation to your de-chirp (U-space) results

Earlier you showed $\text{MAD}_U / \text{MAD}_L$ tightening for IMF2–3 around $\alpha \approx 0.375$ when using the mixed BAO proxy $b_\alpha = \alpha \text{ DH} / r_d + (1-\alpha) \text{ DM} / r_d$:

- Typical ratios $\sim 0.44\text{--}0.57$ (i.e., **43–56%** narrower ridges after de-chirp), plus the bar chart showing tightening across IMFs.
- That's **consistent** with the WL scan: IMF2–3 have a coherent, slow L-dependence with the BAO scale; once you re-parametrize $L \rightarrow U$ via $U(L) \propto \int b(L) / dL$, their instantaneous frequency spreads collapse.

Together, the WL trend and the MAD ratios paint a single picture: **IMF2–3 carry a chirped component whose driver is well captured by a BAO-scale proxy (with a DH-heavy mix), and de-chirping along that driver linearizes the phase evolution.**

Physical reading (quick)

- Negative lock for IMF2–3: as the comoving distance scale **DM/rd or DH/rd increases with redshift**, the **IMF frequency decreases** — i.e., oscillations stretch in

L. That's exactly what a **chirp** looks like in L-space and why the U-space transform tightens the frequency distribution.

- IMF4's sign flip hints at either a **different mechanism** or **edge/aliasing sensitivity**; it's not your primary BAO-linked band.

Recommendations / next steps (fast to run)

1. **Confirm the plateau:** extend WL grid to 0.055, 0.060, 0.070 and check that $|\text{lock}|$ peaks or levels off (especially for IMF2–3).
2. **Edge-safe windows:** when computing windows, drop L within $\pm \text{WL}$ of the BAO support edges; this avoids artificial weakening at large WL.
3. **CI for lock vs WL:** add a block bootstrap (in L) to get 68% bands for each curve, accounting for autocorrelation.
4. **Re-compute U-space at WL = 0.05** using the best $\alpha (\approx 0.375)$ and report **energy-weighted** IMF2–3 tightening (you measured ~ 0.51).
5. **Sign robustness:** compute the same lock metric with **distance correlation** (dCor) as a non-linear check; sign won't apply, but the monotonic rise of strength should persist.

If you want, I can fold the two WL figures + the table and your existing histograms/alpha-sweep into the Word report we built earlier and add a short “WL sensitivity” section.

(Optional) drop-in code: block-bootstrap CI for lock vs WL

```
import numpy as np
import pandas as pd

def block_bootstrap_ci(values, block=75, nboot=1000, seed=11):
    """values ~ array over L-grid; returns 16–84% CI around the mean."""
    rng = np.random.default_rng(seed)
    n = len(values)
    b = max(10, int(block))
    k = int(np.ceil(n / b))
    boots = []
    for _ in range(nboot):
        idx = np.concatenate([rng.integers(0, n-b, size=1) + np.arange(b)
                               for _ in range(k)])
        idx = idx[:n]
        boots.append(np.nanmean(values[idx]))
    lo, hi = np.percentile(boots, [16, 84])
    return float(lo), float(hi)

# example: given your summary rows as a DataFrame "rows" with columns
# ['WL', 'IMF1', 'IMF2', 'IMF3', 'IMF4']
# compute CI on the absolute lock strengths (if you have per-window
# contributions, use those here)
rows = pd.DataFrame({
    "WL": [0.03, 0.035, 0.04, 0.045, 0.05],
```

```

    "IMF1": [-0.050913, -0.044921, -0.039372, -0.045422, -0.053482],
    "IMF2": [-0.068552, -0.071515, -0.083078, -0.105136, -0.114618],
    "IMF3": [-0.048556, -0.066169, -0.090185, -0.110862, -0.129916],
    "IMF4": [ 0.061210, 0.076311, -0.072064, -0.065604, -0.025598]
})
rows_abs = rows.copy()
for c in ["IMF1", "IMF2", "IMF3", "IMF4"]:
    rows_abs[c] = rows_abs[c].abs()

# If you have stored per-window correlations per WL, run block_bootstrap_ci
on those arrays.
# Otherwise (coarse CI), approximate by resampling L with replacement and
recomputing lock per WL.

```

Conclusion

- The **locking metric grows steadily with WL** for IMF2–3 and remains **negative**, reinforcing the interpretation that IMF2–3 encapsulate a **chirped component** whose driver is tightly related to **BAO effective scales**.
- **De-chirp transformations** built from mixed BAO proxies significantly **tighten IMF** ridge frequencies ($\approx 40\text{--}60\%$ for IMF2–3), and the WL scan independently supports that structure (it never weakens through $WL = 0.05$).

Conjecture

The combined evidence (locking sign/strength vs WL, de-chirp tightening, and the α -mix optimum near DH-heavy weight) is consistent with a scenario where a **cosmological length scale that maps closely onto the angular/radial BAO rulers** imprints a **slowly varying phase** on the SN residual field in $L = \log(1+z)$. In that picture:

- IMF2–3 are the **carrier bands** of this chirped modulation,
- IMF4 is either a **secondary component** or **interference/edge-mixing**, and
- the best de-chirp aligns with a **combined radial+transverse** BAO proxy ($\alpha \approx 0.35\text{--}0.40$), hinting that the effective driver is not purely DM/rd nor purely DH/rd but a **blend** related to the anisotropic BAO mapping to distance measures.

If you want these WL plots and the interpretation baked into the Word report (with figure captions and the math above), say the word and I'll produce the .docx with your latest PNGs embedded and labelled.

